Rebecca Turner Woody

Ph.D. Candidate, Astrophysics

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Education

PhD **Harvard University**, Astronomy & Astrophysics — Cambridge, MA
- Advisor: Charlie Conroy — Aug 2021 – present

BS, BA **Johns Hopkins University**, Physics, Mathematics — Baltimore, MD
- Graduated summa cum laude — Aug 2017 – May 2021

Research Interests

Stellar evolution and modeling, stellar ages and isochrone fitting, Galactic archaeology, asteroseismology, convection and mixing length theory, low mass stars, globular clusters

Teaching and Leadership

- Harvard University — Lab Teaching Fellow (Spring 2024): ASTRON 16, Stellar and Planetary Astronomy
- Harvard University — Teaching Fellow (Spring 2022): ASTRON 100, Methods of Observational Astronomy

Honors and Awards

- 2021 Donald E. Kerr Memorial Award, Johns Hopkins University Department of Physics & Astronomy
- 2020 Goldwater Scholar

Talks

- MMT Science Symposium, University of Arizona (April 2026)

Posters

- Searching for r-process Enhanced Metal-poor Stars in the Milky Way Halo, AAS 241 (January 2023)

Workshops

- Yale/CeNAM Nuclear Asteroseismology Workshop (July 2026)
- MESA Summer School, KU Leuven, Belgium (July 2025)
- Binary Stars in the Space Era, Keele University, UK (July 2025)

First-Author Publications

Classical Cases of Non-solar Mixing Length in Low Mass Stars
Woody, Rebecca, Conroy, C., Cargile, P. A., Dotter, A.
(submitted to ApJ)

The Rapid Formation of the Metal-poor Milky Way — 2025
Woody, Rebecca, Conroy, C., Cargile, P., et al.
ui.adsabs.harvard.edu/abs/2025ApJ...978..152W/abstract (ApJ, 978, 152)

The Age–Metallicity–Specific Orbital Energy Relation for the Milky Way's Globular Cluster System Confirms the Importance of Accretion for Its Formation 2021
Woody, Rebecca, Schlaufman, K. C.
ui.adsabs.harvard.edu/abs/2021AJ....162...42W/abstract (AJ, 162, 42)

Co-Author Publications

An Ancient Brown Dwarf Transiting a Metal-poor Thick-disk Star 2026
Adams Redai, J., Chandra, V., Yee, S. W., ..., ***Woody, Rebecca***, et al.
ui.adsabs.harvard.edu/abs/2026ApJ..1002L..41A/abstract (ApJL, 1002, L41)

All-sky Kinematics of the Distant Halo: The Reflex Response to the LMC 2025
Chandra, V., Naidu, R. P., Conroy, C., ..., ***Woody, Rebecca***, et al.
ui.adsabs.harvard.edu/abs/2025ApJ...988..156C/abstract (ApJ, 988, 156)

Distant Echoes of the Milky Way's Last Major Merger 2023
Chandra, V., Naidu, R. P., Conroy, C., ..., ***Woody, Rebecca***, et al.
ui.adsabs.harvard.edu/abs/2023ApJ...951...26C/abstract (ApJ, 951, 26)

A Ghost in Boötes: The Least-Luminous Disrupted Dwarf Galaxy 2022
Chandra, V., Conroy, C., Caldwell, N., ..., ***Woody, Rebecca***
ui.adsabs.harvard.edu/abs/2022ApJ...940..127C/abstract (ApJ, 940, 127)

The Stellar Halo of the Galaxy is Tilted and Doubly Broken 2022
Han, J. J., Conroy, C., Johnson, B. D., ..., ***Woody, Rebecca***, et al.
ui.adsabs.harvard.edu/abs/2022AJ....164..249H/abstract (AJ, 164, 249)

A Tilt in the Dark Matter Halo of the Galaxy 2022
Han, J. J., Naidu, R. P., Conroy, C., ..., ***Woody, Rebecca***
ui.adsabs.harvard.edu/abs/2022ApJ...934...14H/abstract (ApJ, 934, 14)